

ARTYOMOVSK, Russian Federation

WMO: 297720

Lat: 54.3475N

Lon: 93.4350E

Elev: 1572

StdP: 13.88

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.4	(c) -23.4	(d) -34.2	(e) 0.8	(f) -29.1	(g) -28.1	(h) 1.2	(i) -23.2	(j) 15.2	(k) 17.4	(l) 12.1	(m) 12.3	(n) 0.2	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.6	(c) 85.2	(d) 64.9	(e) 81.5	(f) 63.5	(g) 77.9	(h) 62.3	(i) 67.5	(j) 79.4	(k) 65.8	(l) 76.6	(m) 64.2	(n) 73.9	(o) 4.1	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	93.6	71.5	62.1	88.4	69.4	60.8	84.2	68.0	32.9	79.8	31.6	76.4	30.3	73.8	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.1	(b) 11.6	(c) 9.7	(e) -34.4	(f) 90.1	(g) 6.8	(h) 4.2	(i) -39.3	(j) 93.1	(k) -43.3	(l) 95.6	(m) -47.1	(n) 97.9	(o) -52.1	(p) 101.0		
(4) 14.1	(5) 11.6	(6) 9.7	(7) -35.1	(8) 71.3	(9) 6.9	(10) 1.5	(11) -40.0	(12) 72.4	(13) -44.1	(14) 73.3	(15) -47.9	(16) 74.2	(17) -53.0	(18) 75.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	1.2	7.9	22.9	36.2	46.3	59.5	64.2	60.2	47.6	33.7	14.1	5.5
	DBStd	23.67	11.92	11.17	10.24	8.02	7.63	8.06	4.92	5.62	7.18	8.58	12.51	12.51
	HDD50	7226	1513	1178	841	417	165	16	0	2	129	507	1078	1380
	HDD65	11636	1978	1598	1306	865	580	200	74	169	522	971	1528	1845
	CDD50	1169	0	0	0	3	51	300	440	317	57	1	0	0
	CDD65	103	0	0	0	0	1	34	49	19	0	0	0	0
	CDH74	1675	0	0	0	2	78	629	613	320	33	0	0	0
	CDH80	441	0	0	0	0	14	183	168	72	4	0	0	0
Wind	WSAvg	3.0	1.6	2.6	3.4	4.3	4.5	3.6	3.4	3.0	3.0	3.2	2.4	1.6
Precipitation	PrecAvg	29.9	1.1	1.0	1.3	2.0	2.6	3.4	4.2	4.3	3.0	2.6	2.6	1.9
	PrecMax	36.2	1.9	3.0	3.0	3.3	4.6	5.0	7.1	7.4	5.5	7.6	4.4	3.7
	PrecMin	23.7	0.4	0.3	0.4	0.8	0.7	1.7	2.0	2.5	1.0	0.8	1.5	0.5
	PrecStd	3.1	0.4	0.5	0.7	0.7	0.9	0.9	1.0	1.0	1.1	1.2	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.0	36.9	54.7	67.5	81.8	89.5	89.1	86.6	78.7	64.1	42.9	33.6
		MCWB	27.8	31.1	41.2	48.9	58.8	65.7	67.7	66.0	59.7	48.9	38.6	32.2
	2%	DB	25.3	31.3	47.4	61.5	75.1	85.0	85.2	81.6	71.1	55.9	37.4	28.9
		MCWB	23.9	28.5	37.2	46.0	55.5	63.8	66.2	64.4	58.0	45.6	34.2	27.6
	5%	DB	22.1	27.9	42.4	55.8	69.1	81.3	81.0	77.5	65.8	50.1	33.0	26.2
		MCWB	20.7	25.3	34.2	43.2	52.1	62.7	64.5	63.2	54.6	42.7	31.1	25.2
	10%	DB	18.4	24.6	38.5	51.1	63.2	77.1	77.0	72.9	61.0	45.6	30.3	22.8
		MCWB	17.2	22.3	32.5	40.7	50.2	61.1	63.5	61.3	52.5	41.3	28.7	21.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.0	32.2	41.8	50.9	60.1	69.8	70.1	69.0	61.1	50.7	39.6	32.7
		MCDB	28.8	35.2	52.8	64.9	77.8	83.5	84.0	81.1	73.0	59.3	41.5	33.8
	2%	WB	24.1	29.0	38.0	47.3	57.0	66.4	67.8	66.6	58.8	47.1	34.0	27.9
		MCDB	25.1	31.3	46.3	59.3	70.1	79.0	79.6	78.1	70.0	53.7	36.8	28.7
	5%	WB	20.9	25.6	35.0	44.2	54.4	64.5	66.3	64.5	56.2	44.4	31.6	25.2
		MCDB	22.0	27.8	40.9	54.1	64.6	76.4	76.7	74.0	64.0	49.5	33.0	26.2
	10%	WB	17.2	22.2	32.8	41.5	51.6	62.6	64.8	62.6	53.5	41.4	28.8	21.6
		MCDB	18.5	24.5	38.3	50.2	61.2	73.7	74.1	70.2	59.9	45.0	30.4	22.5
Mean Daily Temperature Range	5% DB	MDBR	17.3	21.1	23.4	21.2	22.6	23.4	20.6	21.5	20.7	16.5	15.2	14.9
		MCDBR	15.7	22.5	29.9	31.6	36.5	32.8	30.7	32.5	32.5	27.0	15.3	14.3
		MCWBR	14.7	18.3	20.0	17.6	18.1	14.2	13.2	15.5	17.9	16.3	12.5	13.4
	5% WB	MCDBR	15.5	19.7	26.4	28.9	29.9	27.8	23.9	25.8	28.8	25.1	13.6	13.9
		MCWBR	14.6	16.0	17.4	16.9	16.1	13.1	11.4	13.1	15.7	15.8	11.4	13.1
Clear-Sky Solar Irradiance	taub	0.278	0.313	0.357	0.397	0.408	0.394	0.404	0.393	0.359	0.343	0.315	0.284	
	taud	2.046	2.009	1.953	2.041	2.197	2.355	2.355	2.354	2.391	2.272	2.053	1.979	
	Ebn at Noon	202	235	252	260	265	271	266	261	256	227	181	165	
	Edh at Noon	22	33	44	47	43	37	36	34	29	26	22	19	
All-Sky Solar Radiation	RadAvg	308	633	1054	1347	1585	1879	1676	1395	952	522	266	200	
	RadStd	33	46	102	84	186	194	132	123	119	50	37	25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page